

Broadband virtual private network signalling

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In the corporate voice networks, the virtual private network (VPN) configuration is particularly useful where a shared network infrastructure can be more cost effective and flexible than dedicated leased lines. Current standards for VPN use the international narrowband private signalling system QSIG. Private network signalling standards for broadband networks are also developing rapidly and one can expect that the new emerging signalling systems such as ECMA's BQSIG and ATM Forum's PNNI will play a leading role in the broadband VPN in the future.

1. Introduction

Virtual private networks (VPNs) allow people to make their corporate communications more flexible and cost effective. In the narrowband network, a VPN allows the customers to share their PINX (private integrated service network exchange, also known as PBX) features over geographically distributed sites.

Broadband asynchronous transfer mode (ATM) technology, with even greater bandwidth, will permit integration of the voice and data platforms in addition to providing new services such as videoconferencing and point-to-multipoint calls.

Until then, many issues such as signalling need to be fully addressed before the migration from the narrowband to the broadband VPN is complete. This paper provides an overview of how BQSIG and PNNI, the private broadband signalling systems, support the broadband VPN.

2. What is VPN ?

There has never been a time when the ability to communicate quickly and efficiently within large companies and organisations has been so important. For most domestic and small business users, the telecommunications services provided by the public network operators (PNOs) are adequate for their needs. However, large and geographically distributed corporations, who wish to utilise advanced telecommunications products to improve their business efficiency, find that neither the costly mesh of private connections required to link networks, nor the public ISDN (integrated services digital network) always satisfies their communications requirements.

VPN [1] aims to allow business customers to optimise their communications costs, with respect to dedicated

private networks, by providing private networking services on the public network or on a network set up to offer a set of business services. VPN service providers typically offer a number of 'private' services to customers, including private numbering plan, private service features and advanced billing. An important issue for the service provider is to ensure that these can be delivered across the VPN so that customers receive an 'end-to-end' seamless service.

Figure 1 shows a typical VPN configuration. The customer PINXs are connected to the public ISDN which connects them together and performs private-to-public number translation, via the intelligent network (IN) for instance.

The advantages of implementing corporate telecommunications networks as a VPN apply regardless of geographical location, and may include :

- ability to manage the network,
- guaranteed quality of voice and data,
- improved security,
- customised tariff arrangements,
- a wider range of facilities than those provided by public networks,
- cost-effective geographic connectivity,
- simplified numbering and addressing,
- flexible charging.

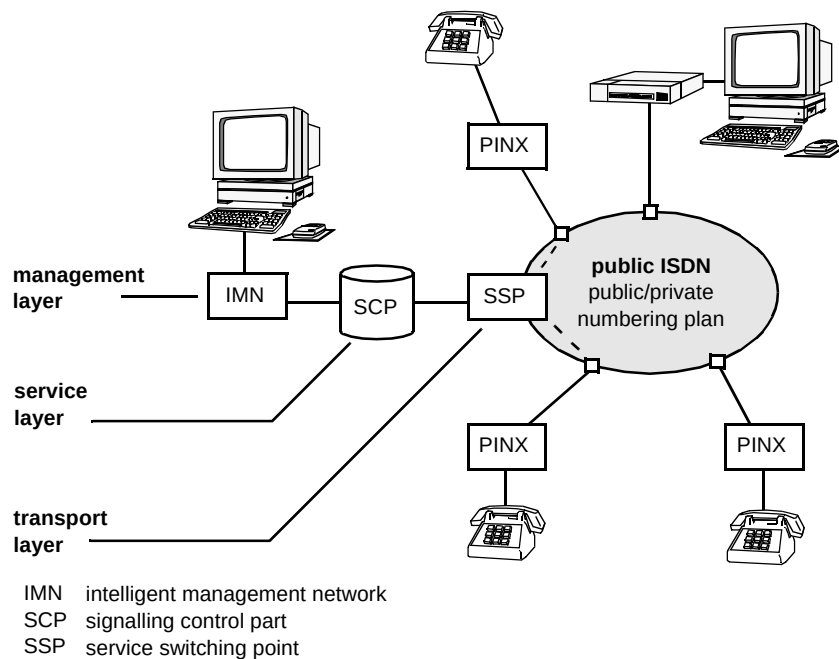


Fig 1 An example of a VPN.

3. VPN architecture

3.1 Public access

User-to-network signalling is used for PINX access to the public ISDN. The International Telecommunications Union — Telecommunications Standardisation Sector (ITU-T) have defined the user/network signalling for ISDN which is called Digital Signalling System No 1 (DSS1) [2] which is the generic name for all three layers of the access signalling system (including the signalling for supplementary services). DSS1 is used for both basic rate (BR) and primary rate (PR) ISDN.

The services are provided in a generic manner and not tailored to the customers as is the case for VPNs. All features are provided by the network and identically available to all the subscribers.

3.2 Private access

Signalling systems have been specified for operation between PINXs (i.e. over the 'Q'-reference point defined by ITU-T). The European version is based on DSS1 and is called QSIG [3]. The standards for QSIG are being developed by ECMA (the European Computer Manufacturers Association). These have been adopted internationally by ISO and this international private network signalling system is called PSS1 (Private Signalling System No 1). QSIG has also been taken from its natural environment to be used in the provision of VPN networks as a VPN access and not only an inter-PINX signalling system.

QSIG is an important signalling system because it supersedes proprietary private signalling and allows migration from them to QSIG. Major PBX manufacturers have implemented QSIG and ensured that it can be mapped to their own proprietary signalling solutions.

Although commonality exists between QSIG and DSS1, there is a significant difference:

- DSS1 is an asymmetrical access protocol of link significance only — there is a customer side and the services are provided by the network side,
- QSIG is a symmetrical network protocol capable of signalling across a network of PINXs/network nodes which means the core network is only used as a transit platform.

3.3 Integrated public/private access

It is generally recognised that most future services provided by the VPN are likely to be implemented using IN and that further service-specific standardisation will be limited. However, IN currently has been standardised around the PSTN and voice services, and it is well recognised that enhancements to the DSS1 standards (as well as other ISDN and IN standards) will be required to fully support services delivered using IN and ISDN.

Standards are currently defining ISDN virtual private networks which include the convergence of public and private network ISDN. Current options under discussion include the mapping and enveloping of QSIG within DSS1,

and the combination of QSIG signalling and DSS1 within a single signalling system (DSS1 enhanced for VPN).

3.4 Network signalling

The main part of the narrowband core network signalling system is referred to as ISUP (ISDN signalling user part) [4]. This system is designed as an international interface for ISDN services. It is used to route the call across the network in conjunction with the IN (when used) to perform number translation for instance between a private and a public number, via INAP (intelligent network application protocol). Such a scenario, including various signalling systems, is represented in Fig 2.

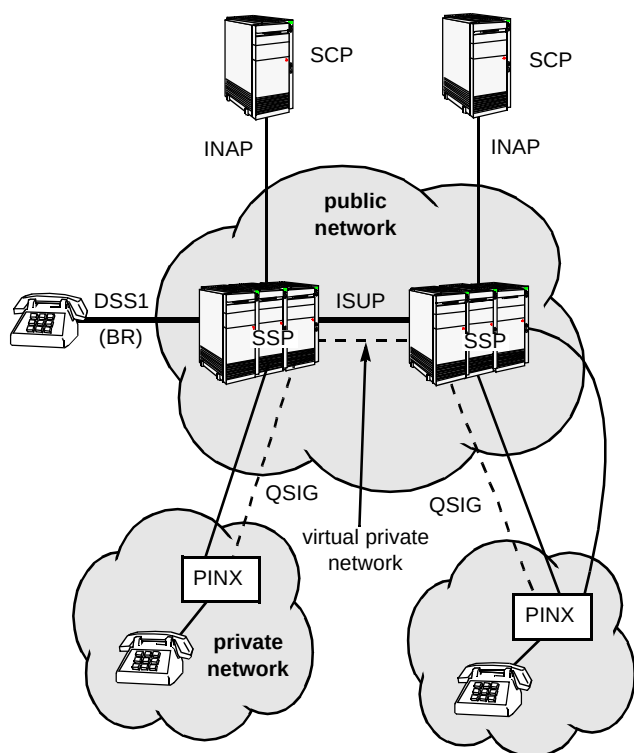


Fig 2 Public and private network architecture.

4. ATM benefits to VPN

4.1 Voice platform evolution to ATM

The crucial competitive factor with ATM is quality of service and pricing — ATM itself is just the technology. Voice over ATM does not necessarily mean less quality — considerable research has been done to show how ATM can reach the quality of voice offered by the narrowband switched networks [5]. What is lost with voice over ATM at present is the range of service features associated with private voice networks.

Theoretically the ATM network could replace the existing dedicated trunk groups interconnecting the narrowband switches, switches to other network

components, and customer sites, as shown in Fig 3. The advantage of the ATM component is that it offers the potential to make more efficient use of the bearer transmission platforms.

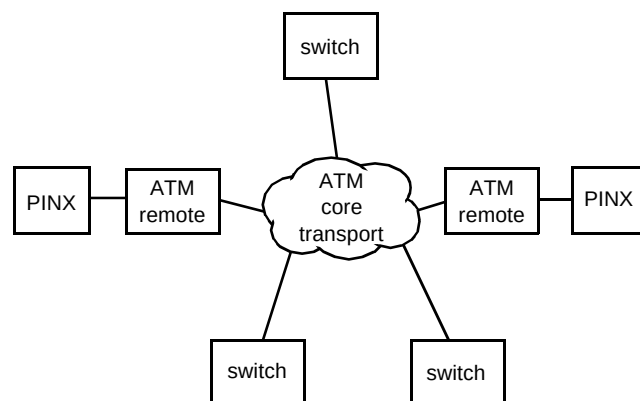


Fig 3 The ATM platform.

In this example, the narrowband PINXs would be connected to the ATM core transport via ATM remote units (or edge switches). The VPN services would be provided by existing narrowband switches interfaced to the ATM transport, or by ATM switches with the narrowband VPN service and signalling.

The narrowband call is set up in a traditional manner using narrowband signalling. Set-up delays can be introduced with the distance between the two terminals.

Constant bit rate is also required as well as high loss priority to guarantee a good quality level.

If a traditional voice call is made, there can be no room for network congestion threatening the conversation. In a voice network, therefore, connections must be delivered with a guarantee. This, then, is the most important aspect which ATM has to prove — its ability to guarantee the quality of service.

It only makes economic sense to change to an ATM infrastructure if it can be demonstrated that:

- a wide ATM base could reduce access distances and offer the opportunity for lower access costs,
- the solution is cost-effective and the network resilient,
- it can offer better voice/data integration.

4.2 Voice/data integration

“For many large organisations, ATM now represents the only cost-effective method for handling data traffic, while future-proofing their network to enable new developments in communications technology

and supporting and developing new routes to their markets.” [6]

As well as offering high-speed files or image transfer, database access, local area network (LAN) interconnection, video communication and co-operative working, a single ATM network can also support existing real-time data services such as Frame Relay, Switched Multi-megabit Data Service (SMDS), and voice [5].

This means that people can customise their traffic according to their needs. Until now they have had to constantly redimension their networks to cope with changing traffic needs.

With the introduction of switched virtual circuits, customer equipment will be able to signal for ATM connections. This will unlock the real power of ATM to set up synonymous customer connections on request. It is essential that broadband VPN (BVPN) and associated signalling support these requirements.

5. Broadband virtual private networks

A broadband VPN is defined as: ‘... a part of a customer’s private network supported by the SVC capability of a B-ISDN switched network’ (i.e. excludes VP tunnelling, leased lines and equivalents) [7–9].

Similar techniques to those used in narrowband VPN technology (for example, using intelligent network number translation) techniques can be applied to broadband networks. By definition, a network without switching would not be considered a VPN — and would require the introduction of switching and translation capabilities to be classed a VPN. A possible BVPN voice network is shown in Fig 4.

The traditional telephony-based network model has long embraced this requirement by defining a number of levels. Data networks have tended to be much smaller and have had less need for this approach. As broadband allows these two worlds to coalesce, the network model may evolve further. A possible BVPN data network is shown in Fig 5.

6. Evolution towards broadband VPNs

6.1 The role of signalling

In the absence of a signalling interface to the network, customers will be restricted to using ‘permanent virtual circuits’ (PVCs). They must contact their network administrator indicating their requirements, such as end-points to be connected, bandwidth required, and service type.

The drawback of this approach is, of course, the time taken and effort involved when the customer wants to change the connection characteristics and increase their bandwidth. In a switched virtual circuit (SVC) environment the situation is quite different [10]. End users may now specify the characteristics of their connections directly, on demand as shown in Fig 6.

To provide any given service to the user all of the interfaces across the network must support the features required by that service in a compatible manner.

The ITU-T Q reference point is still applicable in a broadband network environment, and is recognised as such by ECMA, and similarly by the ATM Forum as their private network/network interface [11]. However, VPNs require additional capabilities to extend private network features beyond their own local domain to the whole of the VPN. The resultant signalling protocol that runs across this interface must be supported by the public network.

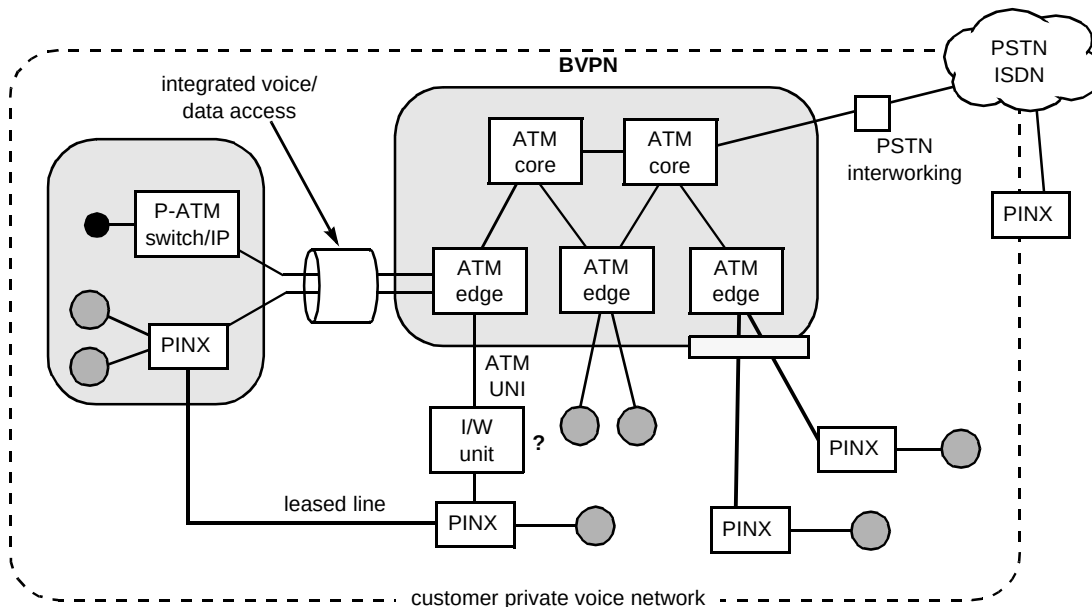


Fig 4 Possible BVPN voice network.

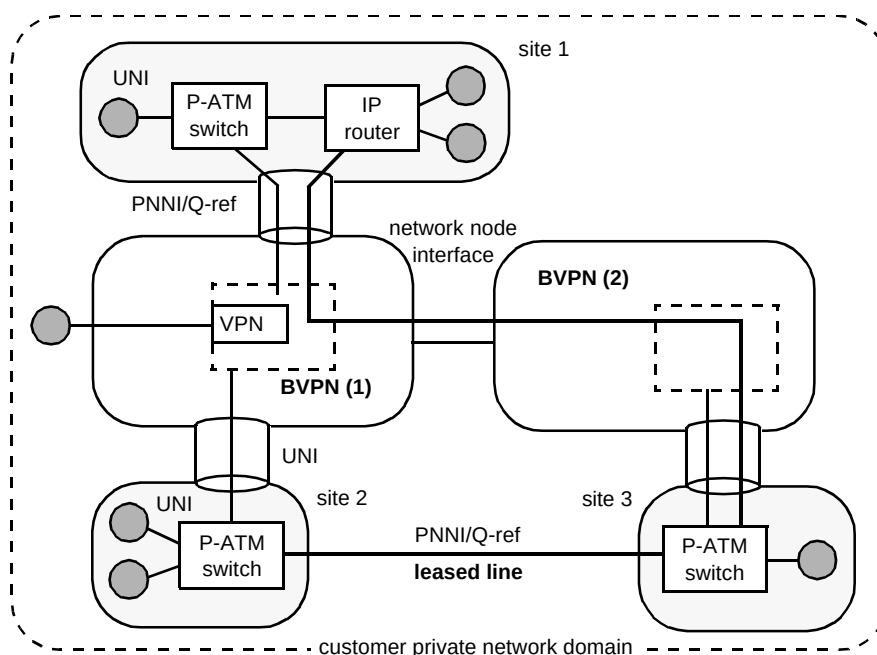


Fig 5 Possible BVPN data network.

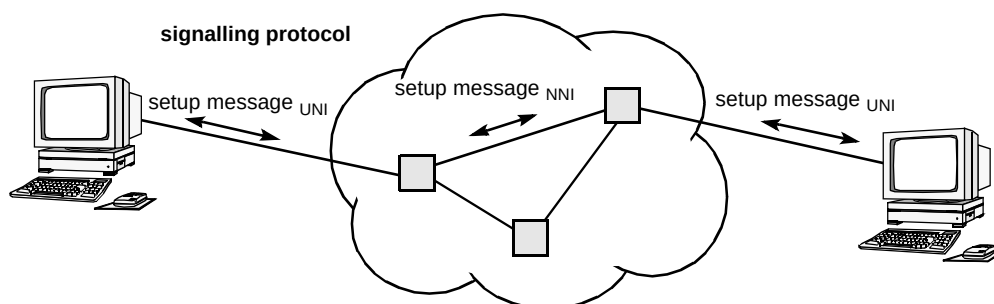


Fig 6 Use of signalling to set up an SVC call.

This is a very simplistic view of the network. The larger the network, the greater is the need for a hierarchy of switching levels, on economic, functional and connectivity grounds.

6.2 Addressing across the public network

In the existing narrowband telephony network, an intelligence 'layer' [12] is used to provide a number translation functions which can be applied to offer various services, e.g. FreeFone, premium rate services, flexible charging. It is expected that the number of core functions will be greatly enhanced in the broadband environment, requiring a more complex and powerful 'intelligent network.'

Any network end-point must be uniquely identifiable within that network [13]. For the public network, therefore, this demands an addressing scheme which offers worldwide unique numbers, such as the E.164 ISDN scheme [14]. However, many private networks will use ATM end-system addressing (AESA) within their networks; E.164 is a public

numbering plan and, as such, numbers are generally only allocated to operators offering a public service. As E.164 numbers are not yet allocated directly to private users for use in private networks, there is a need to translate the customer private number on to an addressing scheme (E.164 or other) that can be used by the network, as shown in Fig 7.

One solution could be provided by PNOs who wish to support all address schemes, maybe as a value added service. This sort of service could be provided by an intelligent network in the future, along with other address manipulation services, such as number portability. However, at the present time, there are no broadband IN standards, and it will be some time before IN technology can be integrated into the broadband area.

Alternatively, operators may choose to use other schemes for VPN addressing, leading to the requirement of global awareness of more than one address scheme if calls are to be successfully routed across multiple public network domains.

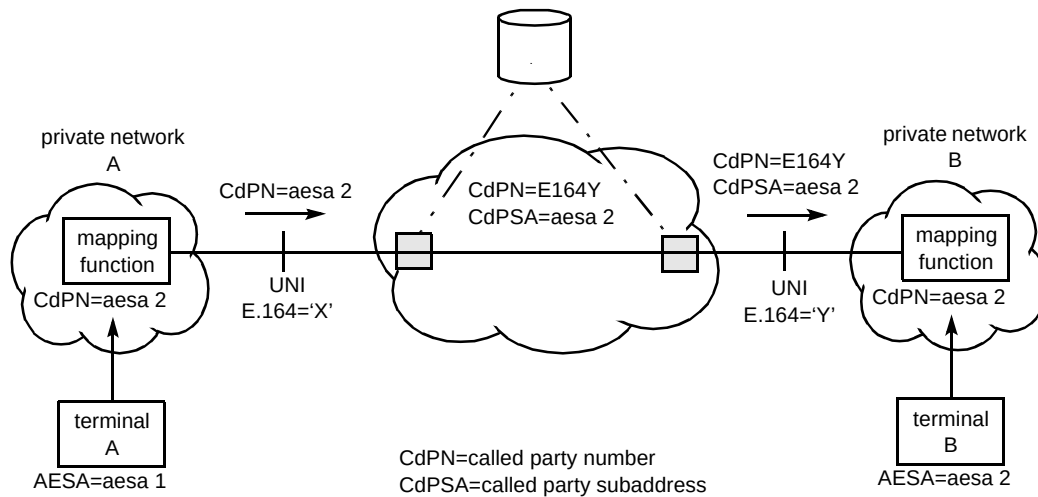


Fig 7 Network addressing.

6.3 Standards for services and signalling in broadband PISNs

Based on experience in developing the QSIG series of standards, ECMA has developed a corresponding signalling system for use in broadband private integrated services networks (B-PISNs) employing ATM technology. B-QSIG (signalling system for use at the Q reference point in broadband networks) operates at the Q reference point between PINXs connected together within a B-PISN employing ATM.

B-QSIG currently comprises the following ECMA standards:

- B-QSIG-GF (generic functional protocol) — extends the basic call/connection control protocol by providing generic support for supplementary services and additional network features,
- B-QSIG-BC (basic call connection/control),
- B-QSIG-SAAL (signalling ATM adaptation layer).

During the development of these standards, ECMA has worked in co-operation with the ATM Forum with a view to achieving alignment, where appropriate, between B-QSIG and the signalling part of the ATM Forum's private network/network interface (PNNI) specification. Whereas the PNNI [11] specification provides automatic configuration and dynamic source routing, B-QSIG is intended for networks that employ static hop-by-hop routing. Therefore B-QSIG-BC particularly allows telecommunications networks with their more static structures and their hop-by-hop routing. Consequently it might become a migration step for such networks towards a fully dynamic and autoconfigured PNNI network.

The future B-QSIG activities will basically go in two directions:

- adaptation of existing QSIG supplementary services to private broadband networks,
- specification of particular support functions for broadband-specific applications.

6.4 Broadband environment

The present capability to provide broadband VPN makes use of the broadband public access, Digital Signalling System No 2 (DSS2) or ATM-Forum UNI 4.0 [13], with the closed user group (CUG) supplementary service.

CUG is defined in ITU-T Recommendation Q.2955.1 and in ATM Forum UNI 4.0 of the supplementary services (B-ISDN). The CUG supplementary services allow users to form groups. A user can be a member of one or more CUGs. Normally, members that belong to a specific CUG can communicate between themselves but not outside the group. Some members can have additional privileges whereby they can be allowed to make outside calls or even receive calls from a member outside the CUG.

In the signalled CUG options provided, the network needs to check the CUG index with the calling party number and then maps the interlock code (IC) — used to check the access rights. Moreover, some members of the VPN might require outside, incoming or both-way access privileges. A table, representing whether a calling party number and IC combination is allowed these privileges, has to be recorded in the network.

Address screening is also an option for providing CUG. It also needs a database of matching calling and called party number pairs. The table has to be constantly updated to accommodate new members. On a small scale, address screening is a better option. However, it is not possible to make a quick transition from address screening to signalled

CUG as there need to be major changes for this to happen. This could be viewed to be very expensive in the long run; however, the table sizes in the address screening option can be considerably cumbersome as the number of members and attributes increases.

In the future, the full SVC will be utilised to provide VPN service in conjunction with PNNI and/or B-QSIG.

6.5 *Broadband VPN support for narrowband VPN*

Section 4.1 described how a narrowband VPN could be overlaid on to an ATM core transport network. However, if the ATM core network supports broadband VPN, it would be possible to support narrowband VPN by interworking the narrowband signalling (for example, QSIG) with the broadband VPN signalling (for example, B-QSIG).

The drawback with this approach is that broadband VPN currently does not support as many supplementary services and features as the narrowband VPN, nor the ability to transparently carry service information between narrowband PINXs.

7. Conclusions

Narrowband VPN already permits customers to share their PINX features across sites using the public network as a transit platform to provide services such as private/public number translation.

Using the ATM technology to provide increased bandwidth and service integration is a great benefit to the user. This paper has described a number of ways in which a VPN can be provided on a broadband ATM network, and the signalling systems used. These are:

- using an ATM core transport network to provide bearer connection capability for existing narrowband VPNs, using current narrowband ISDN signalling,
- using an ATM public broadband ISDN, to provide broadband VPN services, using broadband access signalling,
- using an ATM public broadband ISDN, to provide broadband VPN services, using broadband private network signalling,
- using an ATM public broadband ISDN, to provide narrowband VPN services, by interworking current narrowband signalling with either broadband access or broadband private network signalling.

However, many issues need to be resolved before a truly broadband VPN can be deployed. This paper has highlighted some areas that will have an impact on the signalling systems used for broadband VPN:

- the integration of the IN technology into the broadband network, and convergence between the data and voice platforms,

- the interworking between the narrowband and the broadband signalling systems,
- the addressing schemes to be used to route the calls.

ECMA's B-QSIG and ATM Forum's PNNI will continue to play a leading role in the standardisation of broadband VPN signalling in the future, but compatibility between the different broadband signalling systems is essential.

The broadband VPN will offer many benefits to the corporate networks, including greater bandwidth, flexibility, cost-efficiency, and service integration, in addition to innovative broadband services and capabilities. The current signalling systems provide support for simple broadband VPNs and will continue to progress to support a wider range of VPN services and features.

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He is currently leading the Feature Transparency project for BT, Concert, and the Joint Venture (JV) partners, which aims to provide narrowband Global Virtual Private Network Supplementary Services (for QSIG and DPNSS) transparently over a common platform, independent of the switch manufacturers and is also involved in global signalling network design for BT European JV partners.

He has also been involved in broadband QSIG signalling protocols to represent BT at ECMA TG15 meetings.